ABSHealthcareLite

Architecture Book

Prepared by:

Al Baraka Soft

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ABSHealthcareLite Master Module Dependency Guide

1. Executive Summary

The **ABSHealthcareLite Master Module Dependency Guide** serves as the architectural blueprint for the entire healthcare ERP platform. It defines the structural relationships, data flow, and implementation sequence for Modules 01 through 40.

The platform is designed as a **Multi-tenant SaaS** application. Every business entity is strictly scoped by `CompanyId`. The architecture currently supports **ASP.NET WebForms** and **SQL Server**, with a deliberate, decoupled database design to ensure future portability to **PostgreSQL/Supabase** and application migration to **MVC/Core/Mobile/API** frameworks. The system is inherently multilingual, supporting English, Bangla, Hindi (LTR), and Arabic, Urdu (RTL).

This document ensures that development teams understand which modules must be built first (Foundation), how clinical data flows across departments (Dependencies), and how advanced features (AI, Telemedicine) plug into the core engine.

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2. System Architecture Layers

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+------------------------------------------------------------------------------
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+------------------------------------------------------------------------------
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+------------------------------------------------------------------------------
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+--------------------------------------------------------------------------
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+--------------------------------------------------------------------------
| LAYER 1: PLATFORM SECURITY & GOVERNANCE (Foundation)
| [01] Company/Tenant    [02] User Management     [03] Role & Permission
| [04] Audit Center      [05] Notification        [06] Localization
+--------------------------------------------------------------------------
```

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3. Module Dependency Matrix

Mod	Module Name	Category	Depends On	Used By	Type	Criti
01	Company/Tenant Mgmt	Foundation	None	ALL	Independent	Four
02	User Management	Foundation	01, 03	ALL	Dependent	Four
03	Role & Permission	Foundation	01	02	Dependent	Four
04	Audit Center	Foundation	01, 02	ALL	Dependent	Four
05	Notification Center	Foundation	01	17, 23, 27, 30, 31	Dependent	Four
06	Localization	Foundation	01	ALL	Dependent	Four
07	Branch/Location	Master Data	01	08, 13, 15, 28	Dependent	Four
08	Department Mgmt	Master Data	01, 07	10, 11, 17, 21, 28	Dependent	Four
09	Category Mgmt	Master Data	01	10, 14, 20	Dependent	Four

10	Master Service Catalog	Master Data	01, 08, 09	18, 21, 22, 40	Dependent	Four
11	Doctor Management	Master Data	01, 02, 08	17, 18, 31, 32	Dependent	Four
12	Referral Doctor	Master Data	01	15, 24	Dependent	Four
13	Ward/Cabin/Bed Setup	Master Data	01, 07	28	Dependent	Four
14	Diagnostic Inventory	Master Data	01, 09	21, 22	Dependent	Four
15	Patient Reg & MPI	Patient Journey	01, 06, 07	16, 17, 28, 30	Dependent	Core
16	Patient Profile/Ledger	Patient Journey	15	18, 27, 29, 30	Dependent	Core
17	Appt & Queue Mgmt	Patient Journey	11, 15	18, 31	Dependent	Core
18	Doctor Worklist	Clinical	11, 15, 17	19, 21, 25, 29	Dependent	Core
19	Prescription Mgmt	Clinical	18, 20	26, 30, 32	Dependent	Core
20	Pharmacy Catalog	Clinical	01, 09	19, 26,	Dependent	Core

				29		
21	Sample Collection	Laboratory	10, 15, 18	22	Dependent	Core
22	Result Entry & Analyzer	Laboratory	21, 10	23	Dependent	Adva
23	Result Verification	Laboratory	22, 02, 03	24, 29	Dependent	Adva
24	Report Release	Laboratory	23, 15	30	Dependent	Adva
25	Radiology Workflow	Radiology	10, 15, 18	24, 30	Dependent	Adva
26	MAR	Nursing & IPD	19, 20, 27, 28	29	Dependent	Ente
27	Nursing Station	Nursing & IPD	16, 28	26, 29	Dependent	Ente
28	Bed & Occupancy	Nursing & IPD	13, 15	27, 29	Dependent	Ente
29	Discharge Mgmt	Nursing & IPD	18, 20, 24, 28	30, 31	Dependent	Ente
30	Patient Portal	Engagement	15, 16, 24, 31	32	Dependent	Adva
31	Appt & Follow-Up	Engagement	11, 15, 17, 30	32	Dependent	Adva
32	Telemedicine	Telemedicine	18, 19, 30, 31	None	Dependent	Ente
40	AI Prescription Capture	AI	10, 15, 20	21, 25	Dependent	AI

4. Foundation Modules

Criticality: Foundation These modules form the absolute bedrock of the SaaS platform. No other module can function without them.

- **[01] Company/Tenant Management**: Defines the `CompanyId` boundary.
- **[02] User Management & [03] Role Permission**: Controls access.
- **[04] Audit Center & [05] Notification Center**: Global system utilities.
- **[06] Localization**: Ensures UI and data support EN, BN, AR, UR, HI.
- **[07-14] Master Data**: Branch, Department, Category, Service Catalog, Doctor, Ward Setup, and Inventory. These must be populated before any patient transactions occur.

5. Patient Journey Modules

Criticality: Core These modules handle the entry of the patient into the hospital ecosystem.

- **[15] Patient Registration (MPI)**: Generates the unique Patient ID.
- **[16] Patient Profile Ledger**: The 360-degree view of the patient.
- **[17] Appointment & Queue**: Basic OPD scheduling and token generation.
- **[31] Appointment & Follow-Up**: Advanced scheduling, waitlists, and post-discharge continuity tracking.

6. Clinical Modules

Criticality: Core These modules represent the doctor's interaction with the patient.

- **[18] Doctor Worklist & Encounter**: The core EMR screen for complaints and diagnoses.
- **[19] Prescription Management**: Generates the medication plan.
- **[20] Pharmacy & Medication Catalog**: The master drug database driving the prescription.

7. Laboratory Modules

Criticality: Core to Advanced A strict, linear dependency chain governing the LIS (Laboratory Information System).

- **[21] Sample Collection**: Depends on Investigation Orders. Generates Barcodes.

- **[22] Result Entry & Analyzer**: Depends on Sample Receiving.
- **[23] Result Verification & Approval**: Depends on Result Entry. Enforces clinical governance.
- **[24] Report Release & Delivery**: Depends on Approval. Delivers to Patient Portal.

8. Radiology Modules

Criticality: Advanced

- **[25] Radiology Workflow & Imaging**: Parallel to the Lab workflow but specific to imaging modalities (X-Ray, CT, MRI). Depends on Orders and drives Scheduling, Acquisition, Dictation, and Release.

9. Nursing & IPD Modules

Criticality: Enterprise Highly interdependent modules managing inpatient care.

- **[28] Bed, Ward, Room & Occupancy**: The physical location and billing clock of the patient.
- **[27] Nursing Station & Task Mgmt**: The operational hub. Depends on Bed Occupancy.
- **[26] Medication Administration Record (MAR)**: The execution of the Prescription [19]. Depends heavily on Nursing Station [27] and Pharmacy [20].
- **[29] Discharge Management**: The complex exit workflow. Depends on clearances from Nursing [27], Pharmacy [20], Billing, and triggers Bed Release in [28].

10. Billing & Financial Modules

(Note: Dedicated Finance modules are slated for future documentation, but billing touchpoints are heavily integrated into [10] Service Catalog, [28] Occupancy, and [29] Discharge).

11. Patient Engagement Modules

Criticality: Advanced

- **[30] Patient Portal & Self Service**: The digital front door. Depends on almost all core modules to display Reports [24], Prescriptions [19], and Appointments [31].

12. Telemedicine Modules

Criticality: Enterprise

- **[32] Telemedicine & Virtual Consultation**: The culmination of the outpatient journey. Depends on Portal [30] for access, Appointments [31] for scheduling, and Encounter [18]/Prescription [19] for clinical execution.

13. AI Modules

Criticality: AI

- **[40] AI Prescription Capture**: An intelligent overlay. Depends on the Master Service Catalog [10] and Pharmacy Catalog [20] to map unstructured image data into structured orders.

14. Future Modules

To complete the ERP, future modules will include:

- Finance & Accounts (Vouchers, Ledgers, P&L)
- HR & Payroll
- Fixed Asset Management
- CRM & Marketing
- Advanced Pharmacy POS & Procurement

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15. Recommended Development Dependency Flow

To prevent development blockers, the engineering team MUST build the modules in the following phased sequence:

```
PHASE 1: PLATFORM FOUNDATION (The Bedrock)
[01 Company] ---> [02 User] ---> [03 Roles] ---> [04 Audit] ---> [06 Localizatio
       |
       v
PHASE 2: MASTER DATA (The Dictionaries)
[07 Branch] ----> [08 Dept] ---> [11 Doctor]
```

```
      |               |
      +-> [13 Wards]  +-> [10 Service Catalog]
      |
      +-> [09 Category] -> [20 Pharmacy Catalog] & [14 Diag Inventory]

================================================================================

PHASE 3: PATIENT IDENTITY & OPD (The Front Desk)
[15 Patient Reg] ---> [16 Patient Profile]
      |
      v
[17/31 Appointments] ---> [18 Doctor Encounter] ---> [19 Prescription]

================================================================================

PHASE 4: DIAGNOSTICS (The Lab & Rad Engines)
[18 Encounter Orders]
      |
      +---> [21 Sample Collection] -> [22 Result Entry] -> [23 Verification]
      |                                                            |
      +---> [25 Radiology Workflow] ------------------------------+
                                                                   |
                                                                   v
                                                         [24 Report Release]

================================================================================

PHASE 5: INPATIENT OPERATIONS (The Ward)
[15 Patient Reg] ---> [28 Bed Occupancy]
                           |
                           v
                 [27 Nursing Station] <---> [19 Prescription]
                           |                       |
                           v                       v
                  [26 MAR (Admin Record)] <--------+
                           |
                           v
                 [29 Discharge Management] ---> (Releases Bed in [28])

================================================================================

PHASE 6: DIGITAL HEALTH & AI (The Edge)
[24 Reports] & [19 Prescriptions] & [31 Appointments]
                           |
                           v
                 [30 Patient Portal] <======> [32 Telemedicine]
                           ^
```

```
                   |
        [40 AI Prescription Capture] (Feeds orders back to Phase 4)
```

Cross-Module Relationship Rules

1. **No Circular Dependencies**: Data flows downstream. For example, Lab Result Entry [22] depends on Sample Collection [21], never the reverse.
2. **CompanyId Enforcement**: Every module in Phases 2-6 must filter all database queries by the `CompanyId` established in Phase 1.
3. **Soft Delete Only**: Modules in Phase 4 and 5 (Clinical/IPD) rely on historical data. Master data (Phase 2) cannot be physically deleted (`IsActive = 0` only) to prevent breaking clinical audit trails.

ABSHealthcareLite Development Sequence Guide

1. Executive Summary

The **ABSHealthcareLite Development Sequence Guide** is the official implementation roadmap for engineering teams, architects, project managers, and AI coding agents.

While the system modules are numbered logically by business domain (e.g., 01 to 40), **development order differs significantly from module numbering**. A hospital ERP is a highly relational system. For example, you cannot build Patient Registration (Module 15) without first building Branch/Location (Module 07), and you cannot build the Medication Administration Record (Module 26) without first building the Pharmacy Catalog (Module 20) and Bed Occupancy (Module 28).

This document defines the critical path to ensure development is never blocked by missing dependencies.

2. Development Principles

All development must adhere to the following core principles:

- **Database First**: Schema design, indexing, and relationships must be finalized before UI development.
- **Stored Procedure First**: Business logic should be encapsulated in SQL Server Stored Procedures (for current ASP.NET WebForms compatibility) while keeping logic modular for future ORM/API migration.
- **Multi-Tenant First**: Every single business table MUST include `CompanyId`. Queries must always filter by `CompanyId`.
- **Security First**: RBAC (Role-Based Access Control) must be enforced at the API/Code-behind level, not just UI hiding.
- **Audit First**: `CreatedBy`, `CreatedOn`, `ModifiedBy`, `ModifiedOn`, and `IsActive` are mandatory. Historical data is never physically deleted.
- **Localization First**: No hardcoded UI text. All labels, alerts, and dropdowns must map to resource files supporting English, Bangla, Hindi (LTR) and Arabic, Urdu (RTL).

- **API Ready Design**: Backend logic must be decoupled from WebForms UI events to allow seamless transition to MVC/Core and Mobile APIs.
- **AI Ready Design**: Data must be structured, normalized, and cleanly typed to support future machine learning and LLM integrations.

3. MVP Development Path (Minimum Viable Hospital)

This path delivers a functional OPD and basic Laboratory system.

- **Phase 1 (Foundation)**: Company (01), Users (02), Roles (03), Audit (04), Localization (06).
- **Phase 2 (Master Data)**: Branch (07), Dept (08), Category (09), Service Catalog (10), Doctor (11).
- **Phase 3 (Patient & OPD)**: Patient Reg (15), Appointment (17).
- **Phase 4 (Clinical)**: Doctor Worklist (18), Pharmacy Catalog (20), Prescription (19).
- **Phase 5 (Billing)**: *Core Billing (Future Module)*.
- **Phase 6 (Lab Core)**: Sample Collection (21), Result Entry (22), Report Release (24).

4. Standard Hospital Development Path

This path adds Inpatient Department (IPD) capabilities to the MVP.

- **Phase 1-6**: Same as MVP.
- **Phase 7 (IPD Setup)**: Ward/Bed Setup (13), Bed Occupancy (28).
- **Phase 8 (Nursing)**: Patient Profile (16), Nursing Station (27).
- **Phase 9 (IPD Clinical)**: MAR (26), Discharge Management (29).

5. Enterprise Hospital Development Path

This path adds advanced clinical governance, patient engagement, and AI.

- **Phase 1-9**: Same as Standard Hospital.
- **Phase 10 (Advanced Diagnostics)**: Result Verification (23), Radiology Workflow (25).
- **Phase 11 (Patient Engagement)**: Patient Portal (30), Appt & Follow-Up (31).
- **Phase 12 (Virtual Care & AI)**: Telemedicine (32), AI Prescription Capture (40).

6. Diagnostic Center Only Path

Optimized for standalone pathology labs and imaging centers (skips IPD/Nursing).

- **Phase 1 (Foundation)**: 01, 02, 03, 04, 06.
- **Phase 2 (Master Data)**: 07, 08, 09, 10, 11, 12 (Referral Doctor), 14 (Diag Inventory).
- **Phase 3 (Patient)**: 15, 17.
- **Phase 4 (Billing)**: *Core Billing*.
- **Phase 5 (Laboratory)**: 21, 22, 23, 24.
- **Phase 6 (Radiology)**: 25.
- **Phase 7 (Portal)**: 30.

7. Development Dependency Table

Module	Name	Build Priority	Depends On	Recommended Sprint
01	Company/Tenant Mgmt	1 - Highest	None	Sprint 1
02	User Management	1 - Highest	01, 03	Sprint 1
03	Role & Permission	1 - Highest	01	Sprint 1
04	Audit Center	1 - Highest	01, 02	Sprint 1
06	Localization	1 - Highest	01	Sprint 1
07	Branch/Location	2 - High	01	Sprint 2
08	Department Mgmt	2 - High	01, 07	Sprint 2
09	Category Mgmt	2 - High	01	Sprint 2
10	Master Service Catalog	2 - High	01, 08, 09	Sprint 2
11	Doctor Management	2 - High	01, 02, 08	Sprint 2
12	Referral Doctor	2 - High	01	Sprint 2
13	Ward/Cabin/Bed Setup	2 - High	01, 07	Sprint 2
14	Diagnostic Inventory	2 - High	01, 09	Sprint 2
20	Pharmacy Catalog	2 - High	01, 09	Sprint 2
05	Notification Center	3 - Medium	01	Sprint 3
15	Patient Reg & MPI	3 - Medium	01, 06, 07	Sprint 3
16	Patient Profile	3 - Medium	15	Sprint 3

17	Appt & Queue Mgmt	3 - Medium	11, 15	Sprint 3
18	Doctor Worklist	4 - Medium	11, 15, 17	Sprint 4
19	Prescription Mgmt	4 - Medium	18, 20	Sprint 4
21	Sample Collection	5 - Medium	10, 15, 18	Sprint 5
22	Result Entry	5 - Medium	21, 10	Sprint 5
23	Result Verification	5 - Medium	22, 02, 03	Sprint 5
24	Report Release	5 - Medium	23, 15	Sprint 5
25	Radiology Workflow	6 - Low	10, 15, 18	Sprint 6
28	Bed & Occupancy	7 - Low	13, 15	Sprint 7
27	Nursing Station	7 - Low	16, 28	Sprint 7
26	MAR	8 - Low	19, 20, 27, 28	Sprint 8
29	Discharge Mgmt	8 - Low	18, 20, 24, 28	Sprint 8
30	Patient Portal	9 - Lowest	15, 16, 24, 31	Sprint 9
31	Appt & Follow-Up	9 - Lowest	11, 15, 17, 30	Sprint 9
32	Telemedicine	10 - Lowest	18, 19, 30, 31	Sprint 10
40	AI Prescription	11 - AI	10, 15, 20	Sprint 11

8. Suggested Sprint Planning

Assumes 2-week sprints with a standard full-stack team.

- **Sprint 1**: Foundation & Security (01, 02, 03, 04, 06)
- **Sprint 2**: Master Data Catalogs (07, 08, 09, 10, 11, 12, 13, 14, 20)
- **Sprint 3**: Patient Identity & Basic Scheduling (15, 16, 17, 05)
- **Sprint 4**: Clinical OPD (18, 19)
- **Sprint 5**: Laboratory LIS (21, 22, 23, 24)
- **Sprint 6**: Radiology RIS (25)
- **Sprint 7**: IPD Core (28, 27)
- **Sprint 8**: IPD Clinical & Discharge (26, 29)
- **Sprint 9**: Patient Engagement (30, 31)
- **Sprint 10**: Virtual Care (32)
- **Sprint 11**: AI Integrations (40)
- **Sprint 12**: End-to-End UAT, Load Testing, and Localization Verification.

9. Database Build Sequence

Database schemas must be created in the following strict order to satisfy Foreign Key constraints:

1. **Security & Tenant Tables**: `Company`, `Role`, `User`, `UserRole`.
2. **Location Tables**: `Branch`, `Building`, `Floor`, `Ward`, `Room`, `Bed`.
3. **Master Catalogs**: `Department`, `Category`, `ServiceMaster`, `PharmacyItem`, `Doctor`.
4. **Patient Tables**: `Patient`, `PatientProfile`, `PatientAllergy`.
5. **Encounter Tables**: `Appointment`, `Encounter`, `Prescription`, `PrescriptionItem`.
6. **Lab/Rad Tables**: `SampleCollection`, `LabResult`, `RadiologyStudy`.
7. **IPD Tables**: `BedOccupancy`, `NursingTask`, `MAR`, `DischargeRequest`.
8. **Portal & Telemed Tables**: `PortalAccount`, `TelemedicineSession`.
9. **AI Tables**: `AIPrescriptionCapture`, `AIMapping`.

10. UI Development Sequence

UI development should follow the user's operational flow:

1. **Host Admin Screens**: Tenant creation, global settings.
2. **Tenant Admin Screens**: Masters, users, catalogs.
3. **Front Desk Screens**: Registration, Appointments, Billing.
4. **Doctor Screens**: Worklist, Encounter, Prescription.
5. **Lab Tech Screens**: Sample Collection, Result Entry.
6. **Nursing Screens**: Ward Dashboard, Vitals, MAR.
7. **Patient Screens**: Portal, Telemedicine waiting room.

11. Reporting Development Sequence

Reports should be developed in phases:

1. **Operational Reports**: Daily Registers, Appointment Lists, Pending Lab Queues.
2. **Financial Reports**: Revenue summaries, Collection reports (tied to future billing modules).
3. **Clinical Reports**: Discharge Summaries, Prescriptions, Lab Results.
4. **Analytical/KPI Reports**: Turnaround Times (TAT), Occupancy Rates, AI Accuracy.

12. Mobile App Readiness Sequence

APIs should be developed and exposed in this order:

1. **Patient APIs**: Registration, Login, Book Appointment, View Reports (Supports Module 30).
2. **Doctor APIs**: View Schedule, View Patient Profile, Telemedicine Join (Supports Module 32).
3. **Nursing APIs**: Bedside Vitals Entry, BCMA (Barcode Admin), Task Completion (Supports Modules 26, 27).

13. AI Development Sequence

Module 40 (AI Prescription Capture) should NOT be developed until Sprint 11. *Why?* The AI requires a robust, fully populated `MasterServiceCatalog` (Module 10) and `PharmacyCatalog` (Module 20) to map extracted text against. Furthermore, the downstream billing and lab order generation relies on Modules 21-24 being stable.

14. Recommended Team Structure

For optimal velocity, the engineering team should be structured as follows:

- **1x Solutions Architect**: Database schema design, API contracts, multi-tenant enforcement.
- **2x Backend Developers (C# / SQL)**: Stored procedures, business logic, API endpoints.
- **2x Frontend Developers (ASP.NET / JS / CSS)**: UI/UX, localization implementation, responsive design.
- **1x QA Engineer**: End-to-end testing, RTL layout verification, role-based access testing.
- **1x AI/Integration Engineer (Phase 2)**: OCR models, WebRTC, PACS/DICOM architecture.

15. Final Recommended Roadmap

```
================================================================================
                       ABSHealthcareLite Implementation Roadmap
================================================================================

[ SPRINT 1-2: PLATFORM & MASTERS ]
  +-------------+    +-------------+    +-------------+
  | 01 Company  |--->| 02 Users    |--->| 07 Branches |
  +-------------+    +-------------+    +-------------+
                            |                  |
                            v                  v
                     +-------------+    +-------------+
                     | 03 Roles    |    | 08 Depts    |
                     +-------------+    +-------------+
                                               |
                                               v
                                        +-------------+
                                        | 10 Services |
                                        | 20 Pharmacy |
                                        +-------------+

[ SPRINT 3-4: OPD & CLINICAL ]
  +-------------+    +-------------+    +-------------+
  | 15 Patient  |--->| 17 Appt     |--->| 18 Encounter|
  +-------------+    +-------------+    +-------------+
                                               |
                                               v
```

```
                                        +-------------+
                                        | 19 Prescript|
                                        +-------------+

[ SPRINT 5-6: DIAGNOSTICS ]
  +-------------+    +-------------+    +-------------+    +-------------+
  | 21 Sample   |--->| 22 Results  |--->| 23 Verify   |--->| 24 Release  |
  +-------------+    +-------------+    +-------------+    +-------------+
  | 25 Radiology|------------------------------------------>| (To Portal) |
  +-------------+                                          +-------------+

[ SPRINT 7-8: IPD & NURSING ]
  +-------------+    +-------------+    +-------------+    +-------------+
  | 28 Bed Occ. |--->| 27 Nursing  |--->| 26 MAR      |--->| 29 Discharge|
  +-------------+    +-------------+    +-------------+    +-------------+

[ SPRINT 9-11: ENGAGEMENT, VIRTUAL CARE & AI ]
  +-------------+    +-------------+    +-------------+
  | 30 Portal   |<-->| 31 Follow-Up|<-->| 32 Telemed  |
  +-------------+    +-------------+    +-------------+
         ^
         |
  +-------------+
  | 40 AI Presc | (Feeds structured data into Billing & Diagnostics)
  +-------------+

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```

ABSHealthcareLite System Layer Architecture

1. Executive Summary

The **ABSHealthcareLite System Layer Architecture** defines the complete structural blueprint of the platform. By adopting a strict layered architecture philosophy, the system decouples presentation, business logic, and data access. This ensures that the current ASP.NET WebForms implementation can seamlessly transition to future MVC/Core/API frameworks without rewriting the underlying business rules or database schemas.

This document serves as the master technical blueprint, ensuring that all modules (01-32, 40) adhere to the multi-tenant SaaS model, maintain rigorous data isolation, and provide a scalable foundation for a modern, AI-driven healthcare enterprise.

2. Architectural Principles

- **Multi-Tenant First**: Every architectural layer must respect the `CompanyId` boundary. Data bleeding between tenants is structurally impossible.
- **Security First**: Role-Based Access Control (RBAC) is enforced at the business logic layer, not just the UI.
- **Audit First**: The architecture mandates immutable audit trails (`CreatedBy`, `ModifiedBy`, `IsActive`) across all transactional layers.
- **Localization First**: The UI layer dynamically binds to resource files, supporting LTR (English, Bangla, Hindi) and RTL (Arabic, Urdu) without hardcoded text.
- **API First (Readiness)**: Business logic is encapsulated to allow future RESTful APIs to consume the exact same rules as the WebForms UI.
- **Database Independence**: SQL Server is used now, but schema design avoids proprietary triggers where possible to ensure future PostgreSQL/Supabase portability.
- **AI Ready**: Data is structured and normalized to feed machine learning models (e.g., Module 40).
- **Mobile Ready**: The architecture separates the client layer to support future native iOS/Android applications.

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3. Platform Foundation Layer

The absolute base of the SaaS application.

- **Organization & Company**: Manages the multi-tenant boundaries (Module 01).
- **Branch**: Manages physical locations within a tenant.
- **Configuration**: Global system settings and localization engines (Module 06).
- **Licensing**: Controls feature toggles and tenant subscription limits.

4. Security Layer

Cross-cutting layer protecting all access.

- **Users**: Identity management (Module 02).
- **Roles & Permissions**: RBAC matrix (Module 03).
- **Authentication & Authorization**: Login, session management, and future MFA.
- **Audit**: Centralized tracking of all system actions (Module 04).

5. Master Data Layer

The dictionaries that drive clinical and operational workflows.

- **Departments & Categories**: Organizational taxonomy (Modules 08, 09).
- **Services**: The Master Service Catalog for billing and clinical orders (Module 10).
- **Doctors & Masters**: Provider profiles and referral networks (Modules 11, 12).
- **Inventory/Pharmacy Masters**: Diagnostic and medication catalogs (Modules 14, 20).

6. Patient Management Layer

The core demographic and identity engine.

- **Patient Registration**: Master Patient Index (MPI) generation (Module 15).
- **Patient Identity**: Deduplication and demographic tracking.
- **Patient Profile**: The 360-degree clinical ledger (Module 16).

7. Appointment Layer

The scheduling and patient flow engine.

- **Scheduling**: Doctor rosters, slot management, and waitlists (Modules 17, 31).
- **Follow-Up**: Continuity of care tracking.
- **Queue**: Token generation and waiting area management.

8. Clinical Layer

The physician's workspace.

- **Encounter**: Chief complaints, vitals, and diagnoses (Module 18).
- **Prescription**: Medication plans and e-prescribing (Module 19).
- **Clinical Notes**: Progress notes and structured templates.

9. Diagnostic Layer

The LIS and RIS engines.

- **Laboratory**: Sample collection, result entry, and verification (Modules 21, 22, 23).
- **Radiology**: Imaging workflows, acquisition, and reporting (Module 25).
- **Report Release**: Secure delivery and QR authenticity (Module 24).

10. Pharmacy Layer

Medication dispensing and tracking.

- **Medication Catalog**: Generic/Brand master data (Module 20).
- **Medication Administration**: The nursing MAR execution (Module 26).

11. IPD Layer

Inpatient operations and logistics.

- **Admission & Bed Management**: Occupancy, transfers, and housekeeping (Module 28).
- **Nursing**: Vitals, clinical tasks, and shift handovers (Module 27).

- **Discharge**: Multi-department clearance and continuity of care (Module 29).

12. Finance Layer

The revenue cycle engine (Touchpoints across modules).

- **Billing**: Invoicing for OPD, IPD, Diagnostics, and Pharmacy.
- **Payments**: Cash, credit, and future online gateways.
- **Revenue**: Corporate and insurance tracking.

13. Patient Engagement Layer

The digital front door.

- **Patient Portal**: Self-service booking, report access, and vault (Module 30).
- **Notifications**: SMS, Email, and WhatsApp alerts (Module 05).
- **Messaging**: Secure patient-provider communication.

14. Telemedicine Layer

Virtual care delivery.

- **Virtual Consultation**: Video/Audio integration (Module 32).
- **Virtual Waiting Room**: Remote triage and queue management.
- **Remote Follow-Up**: Digital continuity of care.

15. AI Layer

Intelligent automation.

- **OCR & AI Prescription Capture**: Digitizing unstructured data (Module 40).
- **Smart Mapping**: NLP to map text to the Service Catalog.
- **Recommendation Engine**: Smart billing and duplicate detection.

16. Analytics Layer

Data visualization and KPIs.

- **Dashboards**: Real-time operational views (e.g., Bed Occupancy, Nursing Station).
- **KPIs**: Turnaround times, readmission rates, and revenue metrics.
- **Reporting**: Standardized PDF/Excel exports across all modules.

17. Integration Layer

Future-ready interoperability.

- **APIs**: RESTful endpoints for internal and external consumption.
- **HL7 / FHIR Readiness**: Standardized healthcare data exchange.
- **External Labs & Devices**: Integration with analyzers and smart pumps.

18. Mobile Layer

Client applications.

- **Patient App**: Native extension of the Patient Portal.
- **Doctor App**: Mobile EMR and schedule management.
- **Nursing App**: Bedside vitals and BCMA (Barcode Medication Administration).

19. Future Expansion Layer

Reserved architectural space for specialized modules.

- **Blood Bank**, **ICU**, **OT (Operating Theater)**, **Emergency (ER)**, **Infection Control**, **Public Health**.

---

20. Layer Dependency Diagram

```
+-------------------------------------------------------------------------+
|                        [18] MOBILE LAYER (Apps)                         |
+-------------------------------------------------------------------------+
| [13] PATIENT ENGAGEMENT  | [14] TELEMEDICINE    | [15] AI LAYER         |
+-------------------------------------------------------------------------+
| [07] APPOINTMENT | [08] CLINICAL | [09] DIAGNOSTIC | [11] IPD LAYER     |
+-------------------------------------------------------------------------+
| [06] PATIENT MGMT| [10] PHARMACY | [12] FINANCE    | [16] ANALYTICS     |
```

```
+------------------------------------------------------------------------+
| [17] INTEGRATION LAYER (APIs, HL7, FHIR)                               |
+------------------------------------------------------------------------+
| [05] MASTER DATA LAYER (Catalogs, Doctors, Services)                   |
+------------------------------------------------------------------------+
| [04] SECURITY LAYER (Users, Roles, Audit)                              |
+------------------------------------------------------------------------+
| [03] PLATFORM FOUNDATION LAYER (Company, Branch, Localization)         |
+------------------------------------------------------------------------+
```

21. Cross Layer Data Flow

```
[ PATIENT JOURNEY DATA FLOW ]

(Layer 6: Patient)       [ Registration / MPI ]
                                   |
(Layer 7: Appointment)   [ Schedule / Waitlist ]
                                   |
(Layer 8: Clinical)      [ Encounter / Vitals ]
                                   |
                                   +------------> (Layer 10: Pharmacy) [ Prescrip
                                   |
(Layer 9: Diagnostic)    [ Lab / Rad Orders ]
                                   |
(Layer 15: AI)           [ OCR / Smart Mapping ] <--- (External Rx Upload)
                                   |
(Layer 12: Finance)      [ Billing / Invoicing ]
                                   |
(Layer 11: IPD)          [ Admission / Bed / MAR / Discharge ]
                                   |
(Layer 13: Engagement)   [ Portal Access / Reports / Follow-up ]
```

22. Database Architecture Mapping

Layer	Primary Database Tables (Examples)
Foundation	`Company`, `Branch`, `AppConfiguration`, `LocalizationResource`
Security	`User`, `Role`, `UserRole`, `AuditLog`
Master Data	`Department`, `ServiceMaster`, `Doctor`, `WardMaster`
Patient	`Patient`, `PatientProfile`, `PatientAllergy`
Appointment	`Appointment`, `DoctorSchedule`, `FollowUpPlan`
Clinical	`Encounter`, `ClinicalNote`, `Prescription`
Diagnostic	`LabResult`, `SampleCollection`, `RadiologyStudy`
Pharmacy	`PharmacyItem`, `MedicationAdministrationRecord (MAR)`
IPD	`BedOccupancy`, `NursingTask`, `DischargeRequest`
Finance	`Invoice`, `Payment`, `PatientLedger`
Engagement	`PortalAccount`, `PortalMessage`, `NotificationLog`
Telemedicine	`TelemedicineSession`, `TelemedicineConsent`
AI	`AIPrescriptionCapture`, `AIInvestigationMapping`

---

23. API Architecture Mapping

(Future RESTful API Structure)

Layer	Future API Route (Example)	Purpose
Security	`/api/v1/auth/login`	JWT Token generation
Master Data	`/api/v1/masters/services`	Fetch service catalog
Patient	`/api/v1/patients/{id}`	Fetch patient demographics
Appointment	`/api/v1/appointments/book`	Portal/Mobile booking endpoint
Clinical	`/api/v1/clinical/prescriptions`	Fetch active prescriptions
Diagnostic	`/api/v1/diagnostics/reports/{id}`	Download PDF report
IPD	`/api/v1/ipd/vitals`	Bedside mobile vitals entry
Telemedicine	`/api/v1/telemed/session/{id}`	Fetch WebRTC connection token
AI	`/api/v1/ai/ocr/upload`	Submit image for prescription parsing

---

24. Final Architecture Roadmap

```
=================================================================================
                      ABSHealthcareLite Master Architecture Stack
=================================================================================

[ CLIENT PRESENTATION TIER ]
   +-----------------+   +-----------------+   +-----------------+   +-------------
   | ASP.NET Web UI |   | Patient Portal |   | Mobile App API |   | B2B Integrat
   | (Current)       |   | (Web/Responsive|   | (Future Native)|   | (HL7 / FHIR)
   +-----------------+   +-----------------+   +-----------------+   +-------------

[ BUSINESS LOGIC TIER (C# / Future .NET Core) ]
   +------------------------------------------------------------------------------
   | + AI Prescription Engine (Mod 40)       + Telemedicine Engine (Mod 32)
   | + Clinical Rules Engine (Mod 18,19,26)  + LIS/RIS Engine (Mod 21-25)
   | + IPD & Bed Management (Mod 27,28,29)   + Billing & Finance Engine
   | + Appointment & Queue (Mod 17,31)       + Notification Engine (Mod 05)
   +------------------------------------------------------------------------------
```

```
[ SECURITY & GOVERNANCE TIER ]
   +---------------------------------------------------------------------------
   | + Multi-Tenant Context (CompanyId)      + RBAC Authorization (Mod 03)
   | + Audit Logging (Mod 04)                + Localization Engine (Mod 06)
   +---------------------------------------------------------------------------

[ DATA ACCESS TIER (ADO.NET / Future EF Core) ]
   +---------------------------------------------------------------------------
   | + Stored Procedures (Current)           + ORM Mappings (Future)
   | + Connection Resiliency                 + Read/Write Segregation Readiness
   +---------------------------------------------------------------------------

[ DATABASE TIER (SQL Server / Future PostgreSQL) ]
   +----------------+   +----------------+   +----------------+   +-------------
   | Master Data DB |   | Patient/Cln DB |   | Diagnostics DB |   | Portal/AI DB
   | (Isolated by   |   | (Isolated by   |   | (Isolated by   |   | (Isolated by
   |  CompanyId)    |   |  CompanyId)    |   |  CompanyId)    |   |  CompanyId)
   +----------------+   +----------------+   +----------------+   +-------------

================================================================================
```